

1. **Project Concept:** An interactive fabric world map that serves as a tactile interface for generative climate simulations. It is an 11" x 14" wall-mounted fabric map where each continent is an appliquéd patch with a unique texture.
2. **How to use it:** A user touches a continent, which triggers a nearby display running P5.js to generate a unique "biome" using flow fields (e.g., touching South America creates lush, green swirling particles; touching Antarctica creates slow-moving, crystalline white fractals).
3. **How it is working technically:** The hardware uses an ESP32-S3 DevKitC-1 (which has built-in touch pins) along with conductive fabric and thread. The software side uses the Arduino IDE, P5.js Web Editor, and the p5.serialport library. A key technical challenge involves implementing multi-channel built-in capacitive sensing on the ESP32-S3 without external resistors and integrating Vector Math and Flow Fields (Perlin Noise) in P5.js to create reactive particle systems.

I will now structure this into a concise response.

Action: GENERATE FINAL RESPONSE. The Continental Biome Map is an interactive fabric world map that acts as a tactile interface for generative climate simulations.

- **Project Concept:** It is an 11" x 14" wall-mounted fabric map where each continent is an appliquéd patch, featuring a unique texture (e.g., soft velvet for South America, rough denim for North America).
- **How you use it:** When a user touches a continent, a nearby display generates a unique "biome" using flow fields powered by P5.js. For example, touching the "Amazon" (South America) creates lush, green swirling particles.
- **How it is working technically:** The system uses an ESP32-S3 DevKitC-1 with built-in capacitive touch pins, connected via conductive fabric and thread. The software is developed using Arduino IDE and the P5.js Web Editor, with a focus on implementing multi-channel capacitive sensing





